UNIVERSITY OF BRISTOL

Summer Reassessment for Fall 2024
Computational Neuroscience SEMT30003/4 (Major) Week 6 Exam

FACULTY OF ENGINEERING

SEMT30003
Computational Neuroscience

TIME ALLOWED:
1 hour

This paper contains TWO parts.

[A] The first contains 5 short questions. Each question is worth 2 MARKS. All short questions should be attempted. (10 MARKS)

[B] The second contains 3 in-depth questions. Each of these is worth 10 MARKS. The BEST 2 in-depth questions will be used for assessment. (20 MARKS)

The maximum for this paper is 30 MARKS.

Word counts are hints to help you avoid spending more time than needed. These are not mandatory: You may exceed the advised word count if you wish. You may also use diagrams, illustrations, figures and mathematics as appropriate.

Other Instructions:
Calculators must have the Faculty of Engineering Seal of Approval.

TURN OVER ONLY WHEN TOLD TO START WRITING

Section A: Five short questions—answer all questions

QA.1 [2 marks] Which of the following types of ion channels are typically responsible for propagating the action potential along the axon in humans? (select ALL that apply):

1. Voltage-gated calcium (Ca^{2+}) channels.
2. AMPA ligand-gated sodium (Na^{+}) channels
3. Voltage-gated sodium (Na^{+}) channels.
4. $GABA_A$ chloride (Cl^{-}) channels.
5. Voltage-gated potassium (K^{+}) channels.

Answer: 3,5

QA.2 [2 marks] Which of the following are receptors for *excitatory* neurotransmitters in adult mammals (select ALL that apply):

1. $K_V3.2$
2. AMPA
3. $GABA_A$
4. $Na_V1.1$
5. NMDA

Answer: 2,5

QA.3 [2 marks] Describe the absolute refractory period. Which ion channel(s) is(are) responsible? Which ion(s) is(are) involved, what is their charge, and typical relative concentration(s) inside/outside the cell in mature animals? (≈50–120 words)

Minimal: A period of time following a spike in which a neuron unable to fire further action potentials. It is caused by voltage-gated sodium channels inactivating during the depolarising phase of the spike. When inactive, these channels do not allow positively charged sodium to enter the cell and initiate a spike.

Long: The absolute refractory period is a period of time following a spike in which a neuron unable to fire further action potentials. It is caused by voltage-gated sodium channels inactivating during the depolarising phase of the spike. Spike initiation requires voltage-gated sodium channels to open above a certain voltage threshold, to allow positively charged sodium ions to enter the cell and raise (depolarise) the membrane voltage. When voltage-gated sodium channels are inactive, they do not allow sodium to conduct and therefore prevent the membrane depolarisation required to initiate an action potential.

QA.4 [2 marks] Describe the relative refractory period. Which ion channel(s) is(are) responsible? Which ion(s) is(are) involved, what is their charge, and typical relative concentration(s) inside/outside the cell in mature animals? ($\approx$50–120 words)

Minimal: A period of time following a spike in which a neuron is hyperpolarized and it is more difficult (but not impossible) to initiate another spike. It is caused by inhibitory voltage-gated conductances remaining open for a short while after a spike.

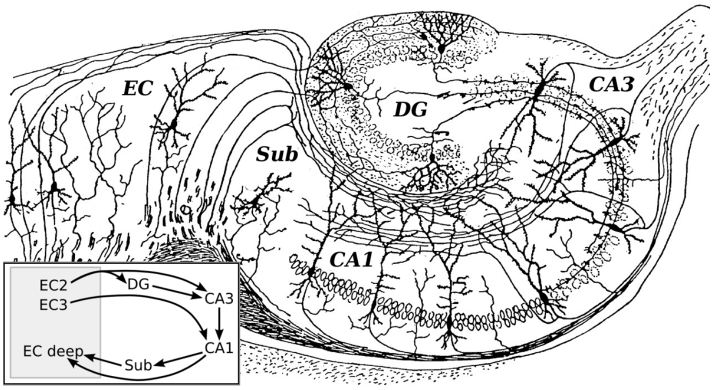
Long: The relative refractory period is a period of time following the action potential where the neuron is inhibited, but sufficiently large depolarising (excitatory) input could still trigger a spike. The relative refractory period is caused by ion channels that open during spiking and which enable conductances that make the neuronal membrane voltage more negative (hyperpolarized, inhibited). Voltage gated potassium channels are the key contributor to this. These channels open during the depolarisation phase of the action potential. They contribute to repolarizing the membrane voltage, and remain open for a period of time (typically 2–5 ms) where they continue to pull the membrane voltage below the resting potential.

QA.5 [2 marks] Like GABA, the neurotransmitter *glycine* is inhibitory in mature animals: When bound to the ionotropic receptor GlyR, glycine lowers the membrane voltage. Like GABA_A receptors, open GlyRs are permiable to chloride ions. What makes the voltage response inhibitory and not excitatory? Explain the role of ionotropic channels and the relative concentration of chloride ions inside versus outside the cell. (≈25-100 words)

Minimal: Glycine is inhibitory when the chloride ion concentration is higher outside the cell. Open Glycine channels allow negatively-charged chloride ions enter, decreasing the membrane voltage.

Long: Ion pumps move chloride ions out of the cell, causing the extracellular concentration of Cl^- to be larger than the intracellular concentration. Entropic forces favour diffusion of Cl^- inward to equalise this concentration difference. Opening the GlyR channel increases the chloride conductance, allowing negatively charged chloride ions to enter the cell and decrease the voltage. The excitatory voltage-gated sodium and calcium channels in neurons open above a certain voltage. Making the membrane voltage more negative makes it harder raise the voltage enough to trigger spikes, and is hence inhibitory. If the concentration gradient were reversed, then glycine would have an excitatory effect, as opening the GlyR channels would allow negative charges to flow out of the cell and raise the membrane voltage.

QA.6 [2 marks] This sketch of the circuity of the hippocampus and surrounding structures is by Santiago Ramón y Cajal. Circle the region that is thought to play an important role in pattern separation and encoding memory representations to send onward to area CA3.



DG and only DG should be circled for full marks. Interpret erased, crossed out answers charitably unless there appears to be a deliberate attempt to present an ambiguous response to game the marking.

In the unlikely event that an over-eager student annotates more than just CA3, for example discussing the entire procedure of encoding, pattern completion, and recall, give full marks only when DG is indicated as the site of densest local recurrent excitatory connections.

Section B: In-depth questions—answer TWO questions

QB.1 This question is about the McCulloch–Pitts neuron and the delta learning rule.

(a) [3 marks] Consider a binary perceptron unit with inputs $\boldsymbol{x} = \{x_1, x_2, \ldots, x_n\}^\top$ and weights $\boldsymbol{w} = \{w_1, w_2, \ldots, w_n\}^\top$, whose output is defined as

$$\hat{y} = \Theta(\textstyle\sum_i w_i x_i) = \Theta(\boldsymbol{w}^\top \boldsymbol{x}), \tag{1}$$

where $\Theta(\cdot)$ is the Heaviside step function (that is, binary threshold)

$$\Theta(r) = \begin{cases} 1 & \text{if } r > 0 \\ 0 & \text{otherwise.} \end{cases}$$

Explain the interpretation of the variables $\boldsymbol{x}$, $\boldsymbol{w}$, and $\boldsymbol{y}$ in terms of both *computation* **and** which aspects of *neuronal signalling and/or anatomy* each corresponds to.

$\boldsymbol{x}$: Each x_i is an input arriving at the neuron, and models (vaguely) the concentration of neurotransmitters released by the presynaptic input at a single incoming synapse.

$\boldsymbol{w}$: Each w_i controls the effect of input x_i on the postsynaptic cell. Larger weights have more effect. Positive weights are excitatory inputs. Negative, inhibitory. Larger weights correspond to larger synapses, or synapses with denser post-synaptic ion channels, or any other biochemical changes at the synapses that increases the impact of input x_i on the dendritic membrane voltage.

y: y is the neurons output. Here, 1 models "spike" and 0 models "no spike". This is a very crude way to capture the all-or-nothing binary nature of the action potential in a computational model.

(b) [3 marks] The delta learning rule for updating weight w_k in (1) is

$$\Delta w_k = \eta \cdot \text{error} \cdot x_k. \tag{2}$$

Re-write Equation (2), replacing error with its expression in terms of $\boldsymbol{w}$ and a single pair of supervised training data $(\boldsymbol{x}, y^*)$, where y^* is the target output.

$$\begin{aligned} \Delta w_k &= \eta \cdot (y^* - \hat{y}) \cdot x_k. \\ &= \eta \cdot [y^* - \Theta(\boldsymbol{w}^\top \boldsymbol{x})] \cdot x_k. \end{aligned} \tag{3}$$

Students who did not substitute to obtain an expression depending on $\boldsymbol{x}$ received partial marks. Some students lost partial marks for omitting the threshold $\Theta(\cdot)$, which is necessary as it has been stated in the definition of the perceptron being considered here (Equation 9).

 Turn Over for QB.1(c)!

(c) [4 marks] Consider the following function,

$$f(\boldsymbol{w};\boldsymbol{x},y^*) = \exp(r) - y^* \cdot r \qquad \text{where } r = \boldsymbol{w}^\top \boldsymbol{x} = \textstyle\sum_i w_i x_i, \tag{4}$$

Suppose we want to minimise $f(\boldsymbol{w};\boldsymbol{x},y^*)$ by adjusting the synaptic weights $\boldsymbol{w}$ (for a given fixed $(\boldsymbol{x},y^*)$.
Show that using gradient descent to change $\boldsymbol{w}$ to minimise Equation (4) leads to a weight update similar to the "delta rule" in Equation (2), if we define $\hat{y} = \exp(r)$.

$$\begin{aligned} \frac{\mathrm{d}}{\mathrm{d}w_i} f(\boldsymbol{w};\boldsymbol{x},y^*) &= \frac{\mathrm{d}}{\mathrm{d}w_i}\exp(r) - y^* \cdot \frac{\mathrm{d}}{\mathrm{d}w_i} r \\ &= [\exp(r) - y^*] \cdot \frac{\mathrm{d}}{\mathrm{d}w_i} r \\ &= [\hat{y} - y^*] \cdot x_i \end{aligned} \tag{5}$$

With gradient descent, we follow the negative of the gradient, so

$$\Delta w = \eta \cdot (-\frac{\mathrm{d}}{\mathrm{d}w_i}) = \eta \cdot [y^* - \hat{y}] \cdot x_i; \tag{6}$$

This is structurally similar to the delta rule in Equation(2).

QB.2 [10 marks] This question is about attractor networks Consider a Hopfield network model:

- Network states are a column vector $\boldsymbol{x} = \{x_1, x_2, \ldots, x_K\}^\top$ for K units ($K = 2$ mostly here) States are either positive or negative one $x_i \in \{-1, 1\}$.
- Network weights are a symmetric matrix $\mathbf{W} \in \mathbb{R}^{K \times K}$. Assume that the weight matrix is set to zero along its diagonal $w_{ii} = 0$.
- Threshold is defined such that non-negative activation triggers a 1 output

$$\Theta(r) = \begin{cases} 1 & \text{if } r \geq 0 \\ -1 & \text{otherwise.} \end{cases}$$

(a) [4 marks] The Hopfield network is an example of a recurrent attractor network. What is an attractor network more generally? How would you expect the states of an attractor network to evolve, starting from a random initialisation? (≈25–50 words)

In the broadest sense, an attractor network system composed of neuron-like units with recurrent connections, and recurrent dynamics that converge to one of several stable fixed points. Some attractor networks, like the Hopfield network, are derived using an ”energy” function. Network dynamics in time (e.g. the Hopfield update rule) are chosen so that they decrease this energy, until reaching a stable local minima.

(b) [3 marks] Write the mathematical expression of the energy of a pattern in the Hopfield network (or the equivalent in words). State whether (remembered) patterns stored in the weight matrix should have higher or lower energy compared to random patterns.

Expect something similar $-\frac{1}{2}\boldsymbol{x}^\top \mathbf{W} \boldsymbol{x}$ or $-\frac{1}{2}\sum_{ij} x_i x_j W_{ij}$. Over-thought answers that are mathematically equivalent in context are also valid. The $\frac{1}{2}$ is optional and no points are lost if it is missing. [+2] points for correct expression and sign or [+1] point for correct expression but incorrect sign.

(c) [3 marks] In a Hopfield network with three neurons there are two patterns to store

$$\boldsymbol{x}_1 = (-1, 1, 1) \text{ and } \boldsymbol{x}_2 = (1, -1, 1);$$

Calculate the weight matrix storing *both* patterns $\boldsymbol{x}_1$ and $\boldsymbol{x}_2$. Use the approach form coursework 1, which calculates the weight matrix for all patterns that are stored in a single step.

$$\boldsymbol{x}_1\boldsymbol{x}_1^\top = \begin{bmatrix} 1 & -1 & -1 \\ -1 & 1 & 1 \\ -1 & 1 & 1 \end{bmatrix}, \qquad \boldsymbol{x}_2\boldsymbol{x}_2^\top = \begin{bmatrix} 1 & -1 & 1 \\ -1 & 1 & -1 \\ 1 & -1 & 1 \end{bmatrix}$$
$$\mathbf{W} = \tfrac{1}{2}(\boldsymbol{x}_1\boldsymbol{x}_1^\top + \boldsymbol{x}_2\boldsymbol{x}_2^\top - 2\mathbf{I}) = \begin{bmatrix} 0 & -1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \tag{7}$$

Full marks for any answer proportional to this (positive scalar factors do not change the dynamics in the Hopfield model we study). Answers correctly solving for weights w_{12} w_{23} w_{13} individually were also valid. Partial marks off for not setting the diagonal to zero (this is stated in the definition of the network being considered, above).

A common mistake involved taking the outer product of the two patterns, e.g. $\boldsymbol{x}_1\boldsymbol{x}_2^\top$ in some form. Some answers included this incorrect approach, but with cancellation of errors leading to partly correct solutions. These could not receive credit.

QB.3 [10 marks] This question is about linear ordinary differential equations used in the leaky integrate-and-fire neuronal model.

(a) [2 marks] You are given a model of a neuron's membrane voltage, whose state v evolves in time according the ordinary differential equation

$$C\dot{v} = g_\ell \cdot (E - v) + I. \tag{8}$$

The variables C, g_ℓ, E, and I are all constants. Re-arrange Equation (8) in the format

$$\tau\dot{v} = v_\infty - v \tag{9}$$

by finding the values of the constants τ and v_∞ in terms of C, g_ℓ, E, and I.

$$\begin{aligned} C\dot{v} &= g \cdot (E - v) + I \\ (C/g_\ell)\dot{v} &= (E + I/g_\ell) - v \\ v_\infty &= E + I/g_\ell \\ \tau &= C/g \\ \tau\dot{v} &= v_\infty - v \end{aligned} \tag{10}$$

Throughout this question, no marks will be deducted for writing v_∞ as $\bar{v}$ or other sensible substitutions with notation in common in computational neuroscience.

(b) [2 marks] In Equation (8), what does the variable g_ℓ represent? Discuss the role of this term in the leaky-integrate-and-fire dynamics, and what we might expect to see if g_ℓ is increased.

The term g_ℓ is the leak conductance. It represents the baseline flow of ions through the membrane. It pulls the neuronal voltage toward the reversal (equilibrium) potential E. A larger leak conductance will make the membrane voltage decay more quickly toward E. It may take a larger current to depolarise (increase) the membrane voltage to the level required to trigger a spike.

(c) [3 marks] If v starts at value v_0 at time $t = 0$, show that Equation (9) has a solution of the form $v(t) = e^{-\alpha t}(v_0 - v_\infty) + v_\infty$ and state the relationship between α and τ.

To show this, students may integrate the ODE using any valid technique, or simply confirm by differentiation or other proof that this form is a correct solution. Here are some examples:

Show equivalent by differentiating to verify:

$$\begin{aligned}
v(t) &= e^{-\alpha t}(v_0 - v_\infty) + v_\infty \\
\dot{v} &= -\alpha\{e^{-\alpha t}v_0 - e^{-\alpha t}v_\infty\} \\
\tfrac{1}{\alpha}\dot{v} &= \underbrace{e^{-\alpha t}(v_\infty - v_0)}_{=-v(t)+v_\infty} \\
\Rightarrow & \\
\tfrac{1}{\alpha}\dot{v} &= v_\infty - v(t), && \text{setting parameters } \alpha = 1/\tau \text{ gives} \\
\tau\dot{v} &= v_\infty - v(t) && \text{recovering the original ODE. } \blacksquare
\end{aligned} \tag{11}$$

Relationship: $\alpha = 1/\tau$.

Variant using undetermined coefficients/ansatz

$$\begin{aligned}
v(t) &= A + Be^{\lambda t} && \text{anstaz} \\
\dot{v} &= B\lambda e^{\lambda t} && \text{differentiate} \\
\dot{v} &= \lambda(v - A) && \text{substitute to get ODE} \\
\tau\dot{v} &= v_\infty - v && \text{match coefficients to given ODE} \\
\dot{v} &= -\frac{1}{\tau}(v - v_\infty) && \lambda = -1/\tau \text{ and } A = v_\infty \\
v(t) &= v_\infty + Be^{-t/\tau} && \text{substitute} \\
v(0) &= v_\infty + Be^{-0/\tau} && \text{apply initial condition to get } B \\
B &= v(0) - v_\infty && \text{solve for } B \\
v(t) &= v_\infty + e^{-t/\tau}(v(0) - v_\infty) && \text{substitute } B \ \blacksquare
\end{aligned} \tag{12}$$

Show equivalent by integrating using integration factor

$$\begin{aligned}
\tau\dot{v} &= v_\infty - v \\
\dot{v} + \tfrac{1}{\tau}v &= \tfrac{1}{\tau}v_\infty && \text{I.F. is } e^{t/\tau} \\
e^{t/\tau}(\dot{v} + \tfrac{1}{\tau}v) &= e^{t/\tau}\tfrac{1}{\tau}v_\infty && \text{Multiply by } e^{t/\tau} \\
e^{t/\tau}v(t) &= \int_{\mathrm{d}t} e^{t/\tau}\tfrac{1}{\tau}v_\infty && \text{Integrate both sides} \\
e^{t/\tau}v(t) &= v_\infty e^{t/\tau} + c && \text{(collect constants to R.H.S.)} \\
v(t) &= v_\infty + ce^{-t/\tau} && \text{Move } e^{t/\tau} \text{ over} \\
v(0) &= v_\infty + ce^{-0/\tau} && \text{evaluate at zero to find } c \\
c &= v_0 - v_\infty && \text{substitute } c \\
v(t) &= v_\infty + e^{-t/\tau}(v_0 - v_\infty) && \text{This matches provided solution. } \blacksquare
\end{aligned} \tag{13}$$

(d) [3 marks] Your colleague wrote computer code to integrate Equation (9) numerically. They used the parameter values $\tau = 10$ ms and $v_\infty = -10$ mV, and initial conditions $v(t = 0) = v_0 = -60$. They used the forward Euler integration method with time step $\Delta t = 5$ ms.

i. [1 marks] What is the order of the error in the Forward Euler method in Δt?

First order

ii. [2 marks] What value for v do you get when calculating $v(t = 5)$ using Forward Euler with $\Delta t = 5$? Does Euler integration over- or under-estimate compared to the true solution of $v(t = 5) \approx -40.32$?

$$\begin{aligned} \dot{v} &= (v_\infty - v)/\tau \\ \dot{v}(0) &= (-10 - -60)/10 = 50/10 = 5 \\ \hat{v}(5) &= v_0 + 5\dot{v}(0) = -60 + 5 \cdot 5 = -60 + 25 = -35 > -44.26 \end{aligned} \tag{14}$$

Forward Euler overestimates (because we have negative curvature, but this detail is not required).